

① $m(a+bx) = a + b \times m(x) \rightarrow \text{Prove}$

$$m(x) = \frac{1}{N} \sum_{i=1}^N x_i, \quad a+bx_1, a+bx_2, \dots, a+bx_N$$

$$m(a+bx) = \frac{1}{N} \sum_{i=1}^N (a+bx_i) \Rightarrow \frac{1}{N} \left(\sum_{i=1}^N a + b \sum_{i=1}^N x_i \right) \Rightarrow \sum_{i=1}^N a = Na \Rightarrow m(a+bx) = \frac{1}{N} \left(Na + b \sum_{i=1}^N x_i \right)$$

$$\Rightarrow a + b \left(\frac{1}{N} \sum_{i=1}^N x_i \right) \Rightarrow \frac{1}{N} \sum_{i=1}^N x_i = m(x) \Rightarrow m(a+bx) = a + b \times m(x) \text{ Hence Proved}$$

② $\text{Cov}(x, a+by) = b \times \text{Cov}(x, y) \rightarrow \text{Prove}$

$$\text{Cov}(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x))(y_i - m(y)) \Rightarrow \text{Cov}(x, a+by) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) [(a+by_i) - m(a+by)]$$

$$\text{Cov}(x, a+by) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) [a+by_i - a - bm(y)] \Rightarrow \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) [b(y_i - m(y))]$$

$$\Rightarrow \frac{b}{N} \sum_{i=1}^N (x_i - m(x))(y_i - m(y)) \Rightarrow \text{Cov}(x, a+by) = b \text{Cov}(x, y) \text{ Hence Proved}$$

③ $\text{Cov}(a+bx, a+bx) = b^2 \text{Cov}(x, x)$ and $\text{Cov}(x, x) = s^2$

$$s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(x))^2 = \text{Cov}(x, x) \Rightarrow \text{Cov}(a+bx, a+bx) = \frac{1}{N} \sum_{i=1}^N [(a+bx_i) - m(a+bx)] [(a+bx_i) - m(a+bx)]$$

$$\Rightarrow m(a+bx) = a + b \times m(x) \Rightarrow \frac{1}{N} \sum_{i=1}^N [a+bx_i - a - bm(x)]^2 \Rightarrow \frac{1}{N} \sum_{i=1}^N b^2 (x_i - m(x))^2$$

$$\Rightarrow \frac{b^2}{N} \sum_{i=1}^N (x_i - m(x))^2 \Rightarrow \text{Cov}(a+bx, a+bx) = b^2 \text{Cov}(x, x)$$

$$\text{Cov}(x, x) = s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(x))^2 \Rightarrow \text{Cov}(a+bx, a+bx) = b^2 s^2$$

$$\therefore \text{Cov}(a+bx, a+bx) = b^2 \text{Cov}(x, x) \text{ and } \text{Cov}(x, x) = s^2$$

④ $g()$: is it always true that $m(g(x)) = g(m(x))$?

Example $\rightarrow x = \{1, 3\}$

non decreasing function $= g(x^2)$ $\Rightarrow m(x) = \frac{1+3}{2} = 2$

$$g(m(x)) = g(2) = 4$$

$$g(x) = \{1^2, 3^2\} \Rightarrow \{1, 9\} \Rightarrow m(x) = \frac{1+9}{2} = 5$$

$$g(m(x)) = g(2) = 4$$

$\therefore m(g(x)) \neq g(m(x))$ (not always true)